

Assignment - 3

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1. Determine the multiplication inverse modulo 26 for each of the following integers
1, 3, ~~5~~, 7, 9, 11, 15, 17, 19, 21, 23, 25

→ (i) 3^{-1}

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$1 = 3 + 2(-1)$$

$$1 = 3[26 + 3(-8)](-1)$$

$$1 = 3 + 26(-1) + 3(-8)$$

$$3(9) + 26(-1) = 1$$

$$\boxed{3^{-1} = 9}$$

(ii) 5^{-1}

$$26 = 5(5) + 1$$

$$26 + 5(-5) = 1$$

$$26 - 5 = 21$$

$$\boxed{5^{-1} = 21}$$

(iii) 7^{-1}

$$26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$26 + 7(-3) + [7 + 5(-1)](-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(-5) + 5(2) = 1$$

$$26 + 7(-5) + [26 + 7(-3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$\boxed{7^{-1} = 15}$$

IV 9^{-1}

$$26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$8 = 26 + 9(-2)$$

$$9 + 8(-1) = 1$$

$$9 + (26 + 9(-2))(-1) = 1$$

$$9 + (26(-1) + 9(2)) = 1$$

$$9(3) + 26(-1) = 1$$

$$\boxed{9^{-1} = 3}$$

V 11^{-1}

$$26 = 11(2) + 4$$

$$4 = 26 + 11(-2)$$

$$11 = 4(2) + 3$$

$$3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$1 = 4 + 3(-1)$$

$$26 + 11(-2) + (11 + 4(-2))(-1) = 1$$

$$26 + 11(-2) + (11(-1) + 4(2)) = 1$$

$$26 + 11(-3) + 4(2) + (26 + 11(-2))(2) = 1$$

$$26 + 11(-3) + 26(2) + 11(4) = 1$$

$$26(3) + 11(7) = 1$$

$$26 - 7 = 19$$

$$\boxed{11^{-1} = 19}$$

VI 15^{-1}

$$26 = 15(1) + 11$$

$$26 + 15(-1) = 11$$

$$15 = 11(1) + 4$$

$$15 + 11(-1) = 4$$

$$11 = 4(2) + 3$$

$$11 + 4(-2) = 3$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$15 + 11(-1) + (11 + 4(-2))(-1) = 1$$

$$15 + 11(-1) + 11(-1) + 4(2) = 1$$

$$15 + 11(-2) + 4(2) = 1$$

$$15 + 11(-2) + (26 + 15(-1))(-2) + (15 + 11(-1))(2) = 1$$

$$15 + 26(-2) + 15(2) + 15(2) + 11(-2) = 1$$

$$15(5) + 26(2) + (26 + 15(-1))(-2) = 1$$

$$15(5) + 26(-2) + 26(-2) + 15(2) = 1$$

$$15(7) + 26(-4) = 1$$

$$\boxed{15^{-1} = 7}$$

VII 17^{-1}

$$26 = 17(1) + 9$$

$$9 = 26 + 17(-1)$$

$$17 = 9(1) + 8$$

$$8 = 17 + 9(-1)$$

$$9 = 8(1) + 1$$

$$1 = 9 + 8(-1)$$

$$26 + 17(-1) + (17 + 9(-1))(-1) = 1$$

$$26 + 17(-1) + 17(-1) + 9(1) = 1$$

$$26 + 17(-1) + 17(-1) + 26 + 17(-1) = 1$$

$$26(2) + 17(-3) = 1$$

$$26 - 3 = 23$$

$$\boxed{17^{-1} = 23}$$

VIII 19^{-1}

$$26 = 19(1) + 7$$

$$7 = 26 + 19(-1)$$

$$19 = 7(2) + 5$$

$$5 = 19 + 7(-2)$$

$$7 = 5(1) + 2$$

$$2 = 7 + 5(-1)$$

$$5 = 2(2) + 1$$

$$1 = 5 + 2(-2)$$

$$19 + 7(-2) + (7 + 5(-1))(-2) = 1$$

$$19 + (26 + 19(-1))(-2) + (26 + 19(-1))(-2) + 5(2) = 1$$

$$19 + 26(-2) + 19(2) + 26(-2) + 19(2) + (19 + 7(-2))(2) = 1$$

$$19(5) + 26(-4) + 19(2) + 7(-4) = 1$$

$$19(7) + 26(-4) + 26(-4) + 19(4) = 1$$

$$19(11) + 26(-8) = 1$$

$$\boxed{19^{-1} = 11}$$

IX 21^{-1}

$$26 = 21(1) + 5$$

$$5 = 26 + 21(-1)$$

$$21 = 5(4) + 1$$

$$1 = 21 + 5(-4)$$

$$21 + (26 + 21(-1))(-4) = 1$$

$$21 + 26(-4) + 21(4) = 1$$

$$21(5) + 26(-4) = 1$$

$$\boxed{21^{-1} = 5}$$

X 23^{-1}

$$26 = 23(1) + 3$$

$$23 = 3(7) + 2$$

$$3 = 2(1) + 1$$

$$\boxed{23^{-1} = 17}$$

XI 25^{-1}

$$26 = 25(1) + 1$$

$$26 + 25(-1) = 1$$

$$26 - 1 = 25$$

$$\boxed{25^{-1} = 25}$$

0	1	2	3	4	5	6	7	8	9	10	11	12	13
A	B	C	D	E	F	G	H	I	J	K	L	M	N
0	1	2	3	4	5	6	7	8	9	10	11	12	13
P	Q	R	S	T	U	V	W	X	Y	Z			

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2. Choose any string of your choice to encrypt using the affine cipher with key values: multiplier 5 and additive 8.

$$C = (P \times K_1 + K_2) \bmod 26$$

$$K_1 = 5, K_2 = 8$$

PT = MALLIKARTJUNA

$$M = (12 \times 5 + 8) \bmod 26 = 68 \bmod 26 = 16 \Rightarrow Q$$

$$A = (0 \times 5 + 8) \bmod 26 = 8 \bmod 26 = 8 \Rightarrow I$$

$$L = (11 \times 5 + 8) \bmod 26 = 63 \bmod 26 = 11 \Rightarrow L$$

$$L = (11 \times 5 + 8) \bmod 26 = 63 \bmod 26 = 11 \Rightarrow L$$

$$T = (8 \times 5 + 8) \bmod 26 = 48 \bmod 26 = 22 \Rightarrow W$$

$$K = (10 \times 5 + 8) \bmod 26 = 58 \bmod 26 = 6 \Rightarrow G$$

$$A = (0 \times 5 + 8) \bmod 26 = 8 \bmod 26 = 8 \Rightarrow I$$

$$R = (17 \times 5 + 8) \bmod 26 = 89 \bmod 26 = 15 \Rightarrow P$$

$$J = (9 \times 5 + 8) \bmod 26 = 53 \bmod 26 = 1 \Rightarrow B$$

$$U = (20 \times 5 + 8) \bmod 26 = 108 \bmod 26 = 4 \Rightarrow E$$

$$N = (13 \times 5 + 8) \bmod 26 = 73 \bmod 26 = 21 \Rightarrow V$$

$$A = (0 \times 5 + 8) \bmod 26 = 8 \bmod 26 = 8 \Rightarrow I$$

CT = QJLLWGIPEVI

3. encrypt "HI" using Affine Cipher with $a=11$ & $b=5$

$$C = (P \times a + b) \bmod 26$$

$$H = (7 \times 11 + 5) \bmod 26 = 82 \bmod 26 = 4 \Rightarrow E$$

$$I = (8 \times 11 + 5) \bmod 26 = 93 \bmod 26 = 15 \Rightarrow P$$

CT = EP

4. The Cipher text "UCR" was encrypted using the affine function $9x + a \bmod 26$. Find the plaintext

$$C = (9P + a) \bmod 26$$

$$a = 9$$

$$P = a^{-1}(C - b) \bmod 26$$

$$b = 2$$

$$9^{-1} = 3$$

$$U = 3(20 - 2) \bmod 26 = 54 \bmod 26 = 2 \Rightarrow C$$

$$C = 3(2 - 2) \bmod 26 = 0 \bmod 26 = 0 \Rightarrow A$$

$$R = 3(17 - 2) \bmod 26 = 4$$

PT = CAT

5. Decrypt the Cipher text using Affine Cipher 'KDYXJ UWDEXZY' using the key 3 multiplier and 4

$$P = a^{-1}(C - b) \bmod 26 \quad a = 3, b = 4$$

$$3^{-1} = 9$$

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$3 + [26 + 3(-8)](-1) = 1$$

$$3 + 26(1) + 3(8) = 1$$

$$26(-1) + 3(9) = 1$$

$$3^{-1} = 9$$

$$26 \div 9 = 2 \text{ R } 8$$

$$K = 9(10 - 4) \bmod 26 = 54 \bmod 26 = 2 \Rightarrow C$$

$$D = 9(3 - 4) \bmod 26 = -9 \bmod 26 = (26 - 9) = 17 \Rightarrow R$$

$$Y = 9(24 - 4) \bmod 26 = 180 \bmod 26 = 24 \Rightarrow Y$$

$$X = 9(23 - 4) \bmod 26 = 171 \bmod 26 = 15 \Rightarrow P$$

$$J = 9(9 - 4) \bmod 26 = 45 \bmod 26 = 19 \Rightarrow T$$

$$V = 9(20 - 4) \bmod 26 = 144 \bmod 26 = 14 \Rightarrow O$$

$$W = 9(22 - 4) \bmod 26 = 162 \bmod 26 = 6 \Rightarrow G$$

$$D = 9(3 - 4) \bmod 26 = -9 \bmod 26 = (26 - 9) = 17 \Rightarrow R$$

$$E = 9(4 - 4) \bmod 26 = 0 \bmod 26 = 0 \Rightarrow A$$

$$X = 9(23 - 4) \bmod 26 = 171 \bmod 26 = 15 \Rightarrow P$$

$$Z = 9(25 - 4) \bmod 26 = 189 \bmod 26 = 17 \Rightarrow R$$

$$Y = 9(24 - 4) \bmod 26 = 180 \bmod 26 = 24 \Rightarrow Y$$

PT = CRYPTOGRAPHY

c) We $K_1 = 13$ (multiplier) $K_2 = 4$ (even though it is not a valid key, encrypt the string "input" and "atten" using the above key and find what is the issue.

→ $C = (P \cdot K_1 + K_2) \bmod 26$ $K_1 = 13, K_2 = 4$
 13 is not co-prime of 26 $\frac{26}{13} = 2$
 so we cannot find inverse of $13 \bmod 26$
 It is not a valid key for affine cipher.

$I \Rightarrow (8 \times 13 + 4) \bmod 26 \Rightarrow 108 \bmod 26 \Rightarrow 4 \Rightarrow E$
 $N \Rightarrow (13 \times 13 + 4) \bmod 26 \Rightarrow 173 \bmod 26 \Rightarrow 17 \Rightarrow R$
 $P \Rightarrow (15 \times 13 + 4) \bmod 26 \Rightarrow 199 \bmod 26 \Rightarrow 17 \Rightarrow R$
 $U \Rightarrow (20 \times 13 + 4) \bmod 26 \Rightarrow 264 \bmod 26 \Rightarrow 4 \Rightarrow E$
 $T \Rightarrow (19 \times 13 + 4) \bmod 26 \Rightarrow 251 \bmod 26 \Rightarrow 17 \Rightarrow R$
 $A \Rightarrow (0 \times 13 + 4) \bmod 26 \Rightarrow 4 \bmod 26 \Rightarrow 4 \Rightarrow E$
 $L \Rightarrow (11 \times 13 + 4) \bmod 26 \Rightarrow 147 \bmod 26 \Rightarrow 17 \Rightarrow R$
 $E \Rightarrow (4 \times 13 + 4) \bmod 26 \Rightarrow 56 \bmod 26 \Rightarrow 4 \Rightarrow E$
 $R \Rightarrow (17 \times 13 + 4) \bmod 26 \Rightarrow 225 \bmod 26 \Rightarrow 17 \Rightarrow R$

ERROR

Both the plain text will produce the same cipher text. The encryption is not one to one and decryption is not possible as 13 is not a valid key.

③ Find the Inverse matrix for the key $\begin{pmatrix} 3 & 2 \\ 5 & 7 \end{pmatrix} \bmod 26$
 $K = \begin{pmatrix} 3 & 2 \\ 5 & 7 \end{pmatrix} \bmod 26$

determinant of key

$$= |3 \times 7 - 2 \times 5|$$

$$= |21 - 10|$$

$$= 11$$

$$11^{-1} \Rightarrow 26 = 11(2) + 4 \quad 4 = 26 + 11(-2)$$

$$11 = 4(2) + 3 \quad 3 = 11 + 4(-2)$$

$$4 = 3(1) + 1 \quad 1 = 4 + 3(-1)$$

$$(26 + 11(-2)) + (11 + 4(-2))(-1) = 1$$

$$26 + 11(2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + 4(2) = 1$$

$$26 + 11(-3) + (26 + 11(-2))(2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19 \quad \boxed{11^{-1} = 19}$$

Adjoint of matrix $(K) = \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26$$

$$K^{-1} = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \bmod 26$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

8) Find the inverse of the following matrix whose entries are considered modulo 26 $\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$

$$\therefore K = \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} \bmod 26$$

$$D(K) = |11 \times 3 - 13 \times 2| = |33 - 26| = 7$$

$$7^{-1} \Rightarrow 26 = 7(3) + 5 \quad \Rightarrow 5 = 26 + 7(-3)$$

$$7 = 5(1) + 2 \quad 2 = 7 + 5(-1)$$

$$5 = 2(2) + 1 \quad 1 = 5 + 2(-2)$$

$$5 + (7 + 5(-1))(-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(-5) + (26 + 7(-3))(2) = 1$$

$$26 + 7(-5) + (26(2) + 7(-6)) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$\boxed{7^{-1} = 15}$$

$$\text{Adj}(K) = \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix}$$

$$K^{-1} = 15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

9 Is the key Valid? $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$

$$\therefore D(K) = |2 \times 21 - 13 \times 14| = |42 - 182| = -140$$

$$= -140 \pmod{26} = 16$$

$$\det K = 16$$

For a key to be Valid its Determinant Value should be Coprime with 26

$\therefore$ It is not a Valid key

10 Encrypt using 2×2 Hill Cipher for the plaintext "SOLVED" using $\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$

$$\rightarrow K = \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \Rightarrow D(K) = |3 \times 11 - 8 \times 7|$$

$$= |33 - 56|$$

$$= -23 + 26 = 3$$

3 is Co-prime

$$\text{SO } L \vee E D \Rightarrow C = K \times P$$

$$\text{SO } \Rightarrow P = \begin{bmatrix} 18 \\ 14 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 54 + 112 \\ 98 + 154 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \pmod{26}$$

$$166 \pmod{26} = 10 \Rightarrow K$$

$$280 \pmod{26} = 20 \Rightarrow U$$

$$L \vee \Rightarrow \begin{bmatrix} 11 \\ 21 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 33 + 168 \\ 77 + 231 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \\ 22 \end{bmatrix}$$

T

W

$$E D = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 12 + 24 \\ 28 + 33 \end{bmatrix} = \begin{bmatrix} 36 \\ 61 \end{bmatrix} \pmod{26} = \begin{bmatrix} 10 \\ 9 \end{bmatrix}$$

K

J

Solved $\Rightarrow$ KUTWKJ

Cipher Text = KUTWKJ